

Algorithm Benchmarking and Outcomes

**Society Connectedness & Health Lab
Yale School of Public Health**

Church Closures Dashboard Project



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Review of the 2023 Format Raw Data

Prepared By: Shelby Golden, M.S. from the Yale School of Public Health Data Science and Data Equity team

Date Created: October 2025

Date Updated: May 19th, 2026

Overview

The project objectives were as follows:

1. **REVIEW:** Evaluate the raw data and previously outlined cleaning, validation, and harmonization methods employed by Dr. Insang Song for accuracy and identify opportunities for improvement.
2. **PREPARATION:** Implement identified improvements to prepare the dataset for analysis and distribution to qualifying collaborators.
3. **METRICS:** Calculate the metrics necessary for dynamic visualization.

Additionally, the project Principal Investigator (PI), Professor Yusuf Ransome, requested the pipeline be reproducible for future collaborators and readily adaptable to potential changes in project scope.

Following receipt of the raw data in Summer 2025, a review was conducted and findings were compared against the original methodology provided by Dr. Song. When an updated dataset was subsequently delivered in May 2026, the pipeline was expanded to accommodate two designated variations: the **2023 Format** and the **2026 Format**.

This document provides an overview of the challenges associated with the 2023 Format data and identifies opportunities for validation and cleaning. Unless otherwise noted, findings outlined here are assumed to apply to the 2026 Format as well; additional details are provided in the comparative review report. This report does not aim to examine every case exhaustively or fully quantify the extent of any issues — rather, it serves to characterize the data at a high level.

Important Considerations

Selected code excerpts are referenced below. The complete code used to generate this report can be found in [“Code/Explore the Raw Data.R”](#).

Ignored Files

Individual-level files are withheld from GitHub and Git tracking in accordance with the Data Use Agreement (DUA) with the data provider. Many of these are stored in `**/KEEP LOCAL/` directories or are explicitly listed under the “Project-Specific Exclusions” section of the `.gitignore` file.

Most files are available on the SOCAH Lab OneDrive for the project, while others are user-specific and do not need to be shared. If there are local-only files for this project, they will be added to the “Church-Closures-Dashboard GitHub LOCAL ONLY Files” directory. This directory will contain `README.md` files specifying the intended use and filepath for each file.

If you do not have access to the SOCAH Lab OneDrive, carefully follow the codebase instructions to establish the repository and replicate results on your own device. Raw data access instructions may vary depending on your institution.

Environment Reproducibility

Results were produced using R (v4.5.2), RStudio (v2026.01.1+403), and Quarto (v1.8.27). The `renv` package (v1.1.4) is included to reproduce the coding environment, capturing all relevant packages and their versions. If issues arise when running the scripts, verify that the environment is initialized and that the correct versions of R, RStudio, and Quarto are being used.

Compatibility

Updating packages, R, or Quarto may introduce compatibility issues with legacy code that will need to be resolved on a case-by-case basis.

First-time users should initialize the environment to install all required packages and ensure a reproducible coding environment. To do so:

1. In the command-line application (i.e., Terminal for Macs or Git Bash for Windows) navigate to the codebase file location.

Command-Line Interface

```
cd "FILE-PATH/Church-Closures-Dashboard/"
```

2. Launch the project by opening the *.Rproj file in RStudio.
3. In the R console, activate the environment by running the following lines of code:

Command-Line Interface

```
renv::restore() # Download packages and their version saved in the
↪ lockfile.
```

i Note

If you are asked to update packages, say no. `renv` is intended to recreate the same environment under which the project was created, making it reproducible. You are ready to proceed when running `renv::restore()` gives the output:

RStudio Console

```
- The library is already synchronized with the lockfile.
```

If you experience any trouble with this step, you might want to confirm that you are using R (v4.5.2) in the RStudio (v2026.01.1+403) and Quarto (v1.8.27). You can also read more about `renv` in their [vignette](#)¹.

Data Dictionary

File name: church_wide_form_071723.csv

Number of rows: 2,601,599

Number of columns: 31

Details were sourced from the Environmental Systems Research Institute (ESRI) documentation on the Data Axle dataset utilized in this study².

- `abi`: American Business Identifier (ABI): The unique identifier for each business, used to associate addresses with a specific organization.
- `year_established`: Indicates the year the business was first established.
- `state`: The state associated with the address.
- `city`: The city associated with the address.
- `zipcode`: The five-digit ZIP code associated with the address.
- `address_line_1`: The primary address line associated with the ABI entry.
- `primary_naics_code`: This dataset combines the six-digit 2022 vintage [North American Industry Classification System \(NAICS\)](#) code and a two-digit proprietary code from Data Axle. The two-digit Data Axle code provides additional classification details not reflected here³.
- `naics8_descriptions`: All entries have NAICS = 813110, indicating they are classified as “Religious Organizations”, which includes churches, shrines, monasteries (except schools), synagogues, mosques, and temples.
- `longitude` and `latitude`: Given as two columns. Provides the geolocation associated with the address listed in that row.
- `2001 to 2021`: One binary column per year (2001–2021) indicating whether a response was recorded for that address: 1 if a response was received, NA if no response was received.

Reduplicate ABI's

Because ABIs are used to uniquely identify entries, we need to ensure that there are no reduplicates.

RStudio Console

```
# Results copied from "Explore the Raw Data.R" file ~ line 147
duplicated(church_wide$abi) %>% table()
```

	Value	Count
1	TRUE	1647336
2	FALSE	954263

It appears that there may be several entries containing reduplicated information, impacting as many as 577727 unique ABIs. We examine each unique ABI that may potentially be reduplicated in the following ways:

1. Confirm that the date binaries are mutually exclusive, indicating that each recorded entry provides unique information.
2. Confirm that the ZIP code remains consistent across entries.
3. Verify that the following metadata fields are consistent across entries: year_established, state, city, primary_naics_code, and naics8_descriptions.
4. A coarse spatial check is applied to confirm that the reported longitude and latitude coordinates for each ABI fall within a maximum difference of one degree. More stringent degree thresholds are investigated in subsequent steps.

Refer to [Appendix - ABI Reduplicates](#) for information on where to find the code that generates these results, as well as the results codebook.

Mutually Exclusive Dates

Each reduplicated ABI reports unique binary records across all year columns. The table below confirms that every column-wise sum per ABI yields 0 or 1 (TRUE), the expected result for unique records.

RStudio Console

```
# Results copied from "Explore the Raw Data.R" file ~ line 242
table(result$Exclusive)
```

	Value	Count
1	TRUE	57727
2	FALSE	0

Moves vs. Reduplications

To appropriately evaluate the remaining three conditions listed above, we need to differentiate variations resulting from explainable sources, such as moves, or errors in data reporting. The primary suspicion is that the detected reduplications arise from alternative addresses associated with the ABIs over a 20-year span.

Table 1 summarizes the possible combinations of the three remaining conditions listed above. For each combination, we estimate the following reasons when the corresponding assumptions are satisfied:

1. LonLat_Similar, Zip_Same, Metadata_Same = TRUE: The church has not moved, no alternative address outside of the reported zip code was used (e.g., PO Box), and the other metadata did not change. This implies that the only variation detected comes from the address_line_1 entry, and no special considerations are required.

RESULTS: Approximately 54% of duplicated entries apply, accounting for 33% of all unique ABIs in the dataset. Of these, nearly 15% used a PO Box at some point, representing almost 5% of all ABIs and 39% of all PO Box entries.

Table 1: Counts and Percentages of Reduplicated ABIs

Counts			Percentages		
Long/Lat = FALSE			Long/Lat = FALSE		
	Zip Code			Zip Code	
Metadata	FALSE	TRUE	Metadata	FALSE	TRUE
FALSE	1,047	9	FALSE	0.18	0.00
TRUE	2	134	TRUE	0.00	0.02
Long/Lat = TRUE			Long/Lat = TRUE		
	Zip Code			Zip Code	
Metadata	FALSE	TRUE	Metadata	FALSE	TRUE
FALSE	49,634	187,570	FALSE	8.59	32.47
TRUE	27,073	309,950	TRUE	4.69	53.65

2. LonLat_Similar, Zip_Same = TRUE and Metadata_Same = FALSE: The church has not significantly moved, which implies that the observed variation is driven by a field other than address_line_1 (e.g., year_established, state, city, primary_naics_code, or naics8_descriptions). These discrepancies may reflect data entry errors or the recording of alternative city locations rather than an actual relocation. Further investigation will ensure completeness.

RESULTS: Approximately 33% of duplicated entries apply, accounting for 20% of all unique ABIs in the dataset. Of these, nearly 12% used a PO Box at some point, representing almost 3% of all ABIs and 19% of all PO Box entries.

3. LonLat_Similar, Metadata_Same = TRUE and Zip_Same = FALSE: This issue can be attributed to two explainable scenarios: either the move was small but resulted in a new zip code, or one of the addresses used a PO Box with a different zip code but the same latitude and longitude as a nearby address. Additionally, there might have been a typographical error in the zip code.

RESULTS: Approximately 5% of duplicated entries apply, accounting for 3% of all unique ABIs in the dataset. Of these, nearly 48% used a PO Box at some point, representing almost 1.4% of all ABIs and 12% of all PO Box entries.

4. LonLat_Similar = TRUE and Zip_Same, Metadata_Same = FALSE: This discrepancy may also be attributable to small relocations that go undetected by the geolocation change threshold yet produce a change in city, zip code, or state for border locations. Variation in the recorded year established may similarly contribute. Both sources of

discrepancy will be further investigated and are expected to be addressed in the same manner as the second or third assumption combinations.

RESULTS: Approximately 9% of duplicated entries apply, accounting for 5.2% of all unique ABIs in the dataset. Of these, nearly 36% used a PO Box at some point, representing almost 1.9% of all ABIs and 15% of all PO Box entries.

5. `LonLat_Similar = FALSE` and `Zip_Same, Metadata = TRUE`: It is unlikely that there would be a significant move without a change in the zip code and other metadata, such as the city or state. It is also possible that non-physical addresses, like PO Boxes, are contributing to this outcome. These are rare occurrences and will be investigated individually to assess for typographical errors.

RESULTS: Approximately 0.02% of duplicated entries apply, accounting for 0.01% of all unique ABIs in the dataset. Of these, nearly 31% used a PO Box at some point, representing almost 0.004% of all ABIs and 0.04% of all PO Box entries.

6. `LonLat_Similar, Zip_Same, Metadata = FALSE`: These entries are most likely associated with a significant move out of the area and will be treated in the same manner as the fifth combination of assumptions. Their zip codes and other metadata might also contain typographical errors that will need to be assessed for completeness.

RESULTS: Approximately 0.18% of duplicated entries apply, accounting for 0.11% of all unique ABIs in the dataset. Of these, nearly 31% used a PO Box at some point, representing almost 0.03% of all ABIs and 0.3% of all PO Box entries.

7. `LonLat_Similar, Metadata = FALSE` and `Zip_Same = TRUE` **OR** `LonLat_Similar, Zip_Same = FALSE` and `Metadata = TRUE`: These results suggest significant relocations with no change in zip code or other metadata. Though a stable zip code is plausible, no change across any address field is unusual. These rare cases will be individually reviewed for typographical errors.

RESULTS: Approximately 0.002% of duplicated entries apply, accounting for 0.001% of all unique ABIs in the dataset. Of these, nearly 91% used a PO Box at some point, representing almost 0.001% of all ABIs and 0.01% of all PO Box entries.

Figure 1 shows the representational distribution across the seven different assumptions listed above. We observe that most outcomes fall into the categories of small or no moves, with over half of the data not indicating any typographical errors. A significant percentage of unique ABIs with at least one PO Box included are also represented in the first two conditions, following the same patterns as the left graph.

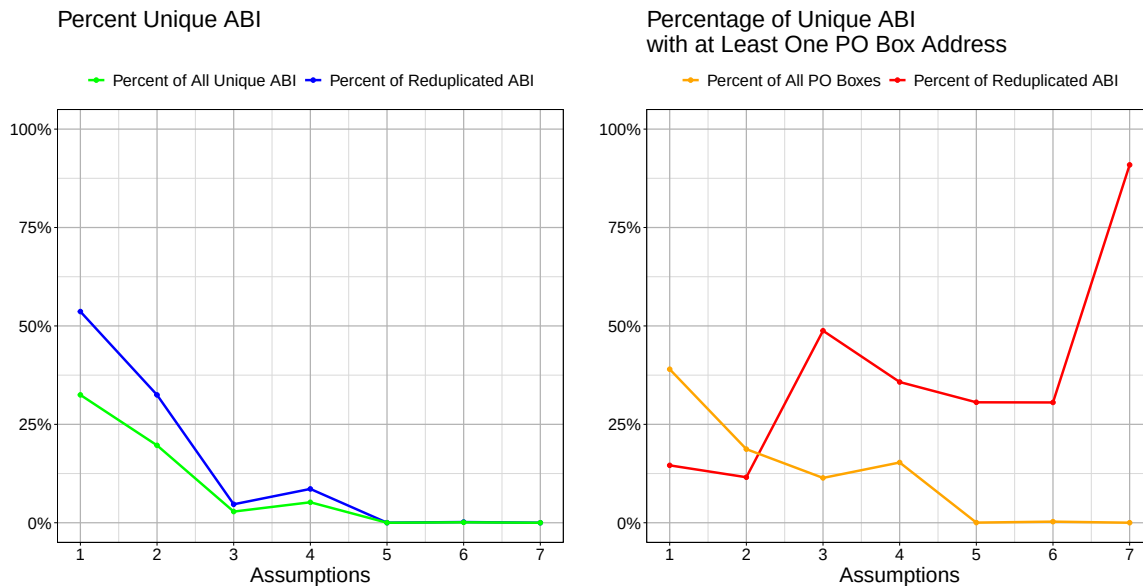


Figure 1: Observed Distributions Across Assumption Conditions

Typographical errors are most likely associated with the the last three assumption combinations. Although most entries appear to satisfy superficial expectations, additional validation is clearly needed to ensure accuracy. PO Box entries also emerge as a notable source of possible errors if excluded or handled the same way as physical addresses.

Closer Look at Assumptions #2

Metadata Fields

Above, we showed that the set of assumptions with `LonLat_Similar`, `Zip_Same` = TRUE and `Metadata_Same` = FALSE, covering 20% of the data, could be associated with typographical errors in the metadata. Other occurrences with `Metadata_Same` = FALSE (assumptions #4, #6, and #7) are likely related to typographical errors not necessarily associated with the metadata fields.

The metadata fields batched together included: `year_established`, `state`, `city`, `primary_naics_code`, and `naics8_descriptions`. All entries have `naics8_descriptions` = "RELIGIOUS ORGANIZATIONS", so this variable will be ignored. Since states and cities are most likely associated together, we will focus on the more granular variable, `city`.

Table 2 tally's whether a specific metadata field varied for a given unique ABI that satisfied the assumptions `LonLat_Similar`, `Zip_Same` = TRUE and `Metadata_Same` = FALSE.

Table 2: Counts and Percentages of ABIs with Varying Metadata

Counts			Percentages		
Year Established = FALSE			Year Established = FALSE		
	NAICS Code			NAICS Code	
City	FALSE	TRUE	City	FALSE	TRUE
FALSE	359	3,490	FALSE	0.19	1.86
TRUE	12,926	130,819	TRUE	6.89	69.74
Year Established = TRUE			Year Established = TRUE		
	NAICS Code			NAICS Code	
City	FALSE	TRUE	City	FALSE	TRUE
FALSE	694	7,360	FALSE	0.37	3.92
TRUE	31,922	0	TRUE	17.02	0.00

Most discrepancies (approximately 70%) are attributable to variation in the reported year established, approximately 17% to NAICS code variation, 4% to city variation (Table 2). The remaining 9% contain a combination of two or all of these factors. Approximately 94.65% of total cases contained an error from either the year established or NAICS Code variable but not city. Approximately 2% contained variation with the city reported and either one or both of the other two variables.

Variation in the reported year established and NAICS code are properties of the business itself rather than of each address under which it was listed. As neither is considered critical to validate completely, both sources of variation will be noted and recorded but not further pursued.

Validating city accuracy poses specific challenges, as approximately 12,000 entries are affected. It is possible that border cities experienced changes in their city association over the three decennial periods captured in this dataset. During the address validation phase, similar addresses differing only by city should be retained as separate entries, with only those validated by the reference source kept.

Refer to [Appendix - Evaluate Metadata in Assumptions #2](#) for information on where to find the code that generates these results, as well as the results codebook.

Zip Codes

Building on the results from Assumption #2, this section further explores the degree of inconsistency in the city reported for a given address. The SimpleMaps Basic [United States Cities Database](#) (version 1.90) is used to identify the recommended city name for a given five-digit ZIP code, and the results are compared against the unique cities listed for each business⁴.

This analysis separates by unique ABI but not unique address, as the focus is on consistency between each unique ZIP code and its corresponding simplemaps result. Granularity beyond the business level is therefore unnecessary. ZIP codes with leading or trailing zeros stripped prior to receipt of this version of the raw data are marked as invalid.

The algorithm returned only one preferred address suggestion per five-digit zip code. NA values were assigned when the zip code was invalid (missing a leading or trailing zero) or when no match was found in the database. Of all businesses, 6.5% had invalid zip codes, while 1.7% had valid zip codes but returned no match. Only 6.7% of these businesses had city variations involving a PO Box.

68% of businesses returned a potential match, while nearly 25% returned no preferred city. The proportion of matches increased as the number of addresses increased (Table 3). There does not appear to be a strong association between PO Box addresses and the absence of a preferred city.

Table 3: Address Count and PO Box by Preferred City Presence

Address Count & Preferred City Presence			PO Box & Preferred City Presence		
Unique-Preferred	Preferred in Unique?		PO Box?	Preferred in Unique?	
	FALSE	TRUE		FALSE	TRUE
1	2,743	7,992	FALSE	22.6	63.3
2	93	92	TRUE	1.3	4.6
3	0	1			

These evaluations indicate inconsistencies in the address entries that warrant validation before they can be considered accurate. Accurate addresses are expected to be more readily matched in the reference database used for geolocation.

Refer to [Appendix - Compare Reported City Against ZIP Code Reference](#) for information on where to find the code that generates these results, as well as the results codebook.

PO Box's

The data processing summary provided by the team noted that all PO Boxes were removed prior to data processing. About 12% of all ABIs listed had at least one PO Box associated with an entry. As shown above, each row-wise entry associated with a given ABI supplies unique information. Therefore, we expect some skew to have been introduced due to this method.

RStudio Console

```
# Detect PO Boxes.
poBox_all <- church_wide[str_which(church_wide$address_line_1, "(?i)PO Box|P
  ↪  O Box"), ]

# Assess prevalence method.
round(length(unique(poBox_all$abi))/length(unique(church_wide$abi))*100,
  ↪  digits = 2)
```

The full extent of the skew is not evaluated on this page. However, to illustrate the issue, two simulated examples based on real instances are provided. These examples highlight two key aspects:

1. A PO Box address is used either before or after other addresses, misrepresenting the span of years open.
2. A PO Box address is used between other addresses for more than three consecutive years, resulting in the incorrect labeling of a church closure.

The second point may introduce skew because the closure detection method defined by Professor Ransome and Dr. Insang identifies a church closure as more than three consecutive zeros between 2001 and 2021. Under this method, sequences such as “10001” and “101” are replaced with “11111” and “111,” respectively, and treated as continuously open. A PO Box used between other addresses that exceeds this threshold would therefore be incorrectly identified as a closure and reopening event.

! Examples are Synthetic

The examples used are synthetic, AI-generated records created to demonstrate structural features of the dataset. Their absence from the actual data was confirmed through city and zip code string matching.

Real examples from the dataset are indexed in the "[Code/Explore the Raw Data.R](#)" file under the section ****ASSUMPTION TO REMOVE PO BOXES****. Note that access to the full dataset is required to view the real examples.

Example #1

Reported year first observed: 2005

Reported year last observed: 2021

Date first closure: N/A

Derived from:

Table 4: Compiled Address Panel Data by Year (2001–2011)

Address	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
1 PO Box 987, Rosewood, PA 19004	1	1	1	1	0	0	0	0	0	0	0
2 123 Maple Street, Rosewood, PA 19001	0	0	0	0	1	1	1	0	0	0	0
3 789 Pine Lane, Rosewood, PA 19003	0	0	0	0	0	0	0	1	0	1	1
4 789 Piine Lane, Rosewood, PA 19003	0	0	0	0	0	0	0	0	1	0	0
5 789 Pine Ln, Rosewood, PA 19003	0	0	0	0	0	0	0	0	0	0	0
6 456 Oak Avenue, Rosewood, PA 19002	0	0	0	0	0	0	0	0	0	0	0

Table 5: Table 7 Continued (2012–2021)

Address	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
1 PO Box 987, Rosewood, PA 19004	0	0	0	0	0	0	0	0	0	0
2 123 Maple Street, Rosewood, PA 19001	0	0	0	0	0	0	0	0	0	0
3 789 Pine Lane, Rosewood, PA 19003	1	1	0	0	0	1	0	0	0	0
4 789 Piine Lane, Rosewood, PA 19003	0	0	0	0	0	0	0	0	0	0
5 789 Pine Ln, Rosewood, PA 19003	0	0	1	1	1	0	0	0	0	0
6 456 Oak Avenue, Rosewood, PA 19002	0	0	0	0	0	0	1	1	1	1

Example #2

Reported year first observed: 2001

Reported year last observed: 2021

Date first closure: 2005 and reopened in 2010

Derived from:

Table 6: Compiled Address Panel Data by Year (2001–2011)

	Address	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
1	321 Chestnut Drive, Greenfield, MA 01301	1	1	1	1	0	0	0	0	0	0	0
2	PO Box 456, Amherst, MA 01002	0	0	0	0	1	1	1	1	1	0	0
3	789 Elm Street, Greenfield, MA 01301	0	0	0	0	0	0	0	0	0	1	1
4	456 Oak Road, Greenfield, MA 01301	0	0	0	0	0	0	0	0	0	0	0

Table 7: Table 9 Continued (2012–2021)

	Address	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
1	321 Chestnut Drive, Greenfield, MA 01301	0	0	0	0	0	0	0	0	0	0
2	PO Box 456, Amherst, MA 01002	0	0	0	0	0	0	0	0	0	0
3	789 Elm Street, Greenfield, MA 01301	1	1	0	0	0	0	0	0	0	0
4	456 Oak Road, Greenfield, MA 01301	0	0	1	1	1	1	1	1	1	1

These examples do not show the geolocation associated with each address. It was observed that PO Box coordinates occasionally coincide with those of a physical address. This is acceptable when the intent is to closely associate the two records; however, it may introduce error when multiple coordinate matches exist or when the PO Box location is sufficiently distant from the physical address to suggest relocation outside of the community.

The full extent of this issue warrants careful handling in the final algorithm. As a preliminary step, a coarse check assessing how much geolocation varies between a PO Box and all other addresses listed under the same ABI is shown below. The goal is to flag large discrepancies that could exacerbate the issue. Note that results should be interpreted cautiously at this stage, as no association between PO Box and physical addresses is assumed.

28% of evaluations revealed that only a PO Box was listed, corresponding to almost 3% of all unique ABI. All such entries are associated with a `Summary Outcome = FALSE` result. Excluding evaluations where only PO Boxes were used, 38% of entries showed at least one PO Box to physical address alignment. NA values were detected in one or more comparators, though these constituted fewer than 1% of the data.

Table 8: Summary Statistics for Failed Longitude and Latitude Values

Longitude Failed				Latitude Failed			
Statistic	Longitude			Statistic	Latitude		
	Min	Mean	Max		Min	Mean	Max
Min.	0.0000	0.0000	0.0000	Min.	0.0000	0.0000	0.0000
1st Qu.	0.0050	0.0060	0.0060	1st Qu.	0.0040	0.0050	0.0050
Median	0.0120	0.0140	0.0150	Median	0.0100	0.0120	0.0130
Mean	0.0370	0.0404	0.0449	Mean	0.0266	0.0292	0.0322
3rd Qu.	0.0290	0.0330	0.0360	3rd Qu.	0.0240	0.0270	0.0290
Max.	30.8260	30.8260	47.4480	Max.	11.0440	11.0440	11.0440

Restricting the analysis to entries where `Summary Outcome = FALSE` and excluding evaluations limited to PO Boxes, the summary statistics indicate that most geolocation differences trend towards zero, with the third quartile for each measure falling below 1 degree. However, some entries exhibit notably large discrepancies, with longitude differences reaching up to 47 degrees (Table 8).

Refer to [Appendix - Impact Assessment of PO Box Removal](#) for information on where to find the code that generates these results, as well as the results codebook.

States Represented

All 50 states are represented, along with D.C. and five U.S. territories. Confirming representation of all cities is not applicable, as some cities may genuinely have no religious institutions rather than reflecting missing data. The following abbreviations represent non-state U.S. jurisdictions that are also present:

- **DC** – District of Columbia
- **PR** – Puerto Rico
- **VI** – U.S. Virgin Islands
- **GU** – Guam
- **MP** – Northern Mariana Islands
- **FM** – Federated States of Micronesia

RStudio Console

```
# Confirm all 50 U.S. states are represented.
unique(church_wide$state) %>% [. %in% datasets::state.abb] %>% length()

# Other state-level codes that are not one of the 50 U.S. states.
unique(church_wide$state) %>% [. %!in% datasets::state.abb]
```

Appendix

ABI Reduplicates

For the implementation, refer to section **AMERICAN BUSINESS IDENTIFIER (ABI) REDUPLICATES** in “Code/Explore the Raw Data.R”. The data processed consisted of all ABI records for which a duplication was identified (i.e., multiple entries sharing the same ABI). Results were written to "Data/Results/KEEP LOCAL/From Explore the Raw Data/ABI Duplicates Test_05.16.2025.csv".

Table 9: Codebook for Evaluation Output Fields

Field	Description
abi	Unique business identifier over which the evaluation is performed.
Exclusive	Boolean. TRUE if date columns sum to 0 or 1, indicating that the information provided by each is mutually exclusive.
Zip_Same	Boolean. TRUE if the zip code is consistent across all addresses associated with a given business ID.
Metadata_Same	Boolean. TRUE if the following metadata fields are consistent across all addresses associated with a given business ID: year_established, state, city, primary_naics_code, and naics8_descriptions.
LonLat_Similar	Boolean. TRUE if the difference between the minimum and maximum longitude AND latitude values are each less than 1 degree.

Evaluate Metadata in Assumptions #2

For the implementation, refer to subsection **Identify Source of Reduplicated Entries** in “[Code/Explore the Raw Data.R](#)”. The data processed were from "Data/Results/KEEP LOCAL/From Explore the Raw Data/ABI Duplicates Test_05.16.2025.csv". Results were written to "Data/Results/KEEP LOCAL/From Explore the Raw Data/Metadata that Should Be the Same_04.10.2026.csv".

Table 10: Codebook for Evaluation Output Fields

Field	Description
abi	Unique business identifier. Evaluation is performed over each unique business ID in the subset that meets the second set of reduplication assumptions.
Year	Boolean. TRUE if all addresses associated with the business ID share a single unique value for <code>year_established</code> .
State	Boolean. TRUE if all addresses associated with the business ID share a single unique value for <code>state</code> .
City	Boolean. TRUE if all addresses associated with the business ID share a single unique value for <code>city</code> .
NAICS	Boolean. TRUE if all addresses associated with the business ID share a single unique value for <code>primary_naics_code</code> .

Compare Reported City Against ZIP Code Reference

For the implementation, refer to subsection **Source of Reduplicated Entries - Assumptions #2** in [“Code/Explore the Raw Data.R”](#). The data processed were from "Data/Results/KEEP LOCAL/From Explore the Raw Data/Metadata that Should Be the Same_04.10.2026.csv". Results were written to "Data/Results/KEEP LOCAL/From Explore the Raw Data/City Name Variation_04.10.2026.csv".

Table 11: Codebook for Evaluation Output Fields

Field	Description
abi	Unique business identifier. Evaluation is performed over each unique business ID in the subset that meets the second set of reduplication assumptions and where the city similarity evaluation did not pass.
Preferred City Name	Preferred address matched to the business using the zip code and the simplemaps database.
# Entries	Number of rows, or possible addresses, associated with each business ID.
PO Box?	Boolean. TRUE if any address associated with the business ID corresponds to a PO Box.
Similar Names	City names in the dataset grouped into clusters of similar names by string similarity.
Unique Names	Distinct city name outcomes present in the dataset after grouping.
Unique-Similar	Difference between the number of unique city names and the number of unique similar city name clusters.
# Cities Suggested	Total number of city matches suggested by the simplemaps database for the associated zip code.
Preferred in Unique?	Boolean. TRUE if the preferred city name returned by the simplemaps database is already present in the dataset.
Unique-Preferred	Difference between the number of unique city names and the number of preferred city names.

Impact Assessment of PO Box Removal

For the implementation, refer to subsection **ASSUMPTION TO REMOVE PO BOX'S** in “Code/Explore the Raw Data.R”. The data were subset to include only ABIs associated with at least one PO Box address at any point in the observation period. Results were written to “Data/Results/KEEP LOCAL/From Explore the Raw Data/PO Box Geolocation_04.10.2026.csv”.

Table 12: Codebook for Evaluation Output Fields

Field	Description
abi	Unique business identifier. Evaluation is performed over each unique business ID associated with at least one PO Box address.
address_line_1	PO Box address. Evaluation is performed over each unique PO Box associated with a given business ID.
Summary Outcome	Boolean. TRUE if any entries sharing the same PO Box number pass both the longitude and latitude similarity tests (threshold: 0.002 degrees each).
n_failed	Number of entries that failed the longitude and latitude similarity tests.
n_total	Total number of occurrences of the PO Box across all entries.
lon_failed_[min mean max]	Minimum, mean, and maximum longitude values among entries that failed the similarity test.
lat_failed_[min mean max]	Minimum, mean, and maximum latitude values among entries that failed the similarity test.

References

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3. U.S. Census Bureau. North american industry classification system (NAICS). <https://www.census.gov/naics/?input=813110&year=2022&details=813110> (2026).
4. SimpleMaps. United states cities database. <https://simplemaps.com/data/us-cities>.